SAMSUNG

MEMCON

Processing In Memory (PIM)

Accelerating AI with Memory

Panel discussion

Mar 29 | 1:00 PM

Seungwon Lee

VP of Technology (Master)
Samsung Electronics

Seungwon Lee (VP of Technology) leads Computing SW Lab (Tech Unit) at SAIT(Samsung Advanced Institute of Technology) of Samsung Electronics.

His research interests include large scale deep learning computing SW and In/Near memory computing SW.
He is leading PIM SW Stack including programming model since 2018.

He had leaded a team for a compiler and quantizer for in-house neural processor from 2016 to 2020. Before neural processor, he had leaded a team for a SDK and programming model for in-house multiple DSPs. Lee received a Ph.D degree in computer science and engineering from Seoul National University.

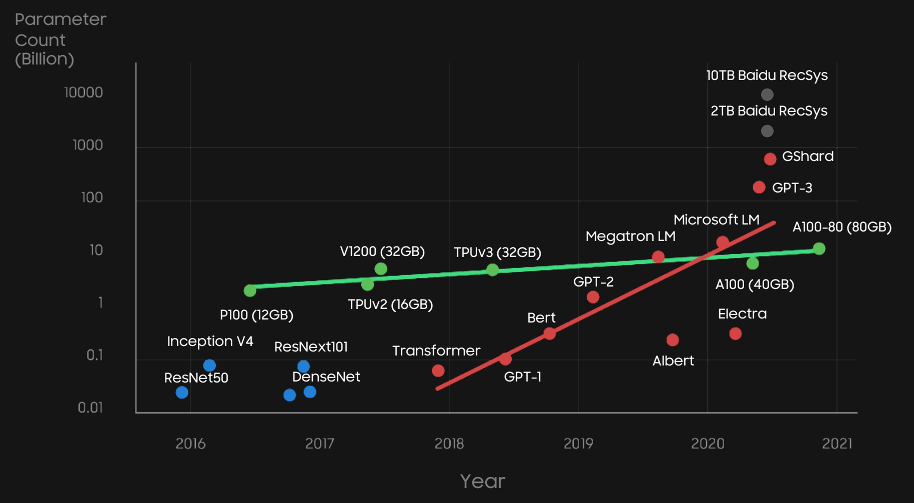
Processing in Memory (PIM)

Video: https://youtu.be/3yMEz45weBA

Significant data increase comes from AI such as ChatGPT

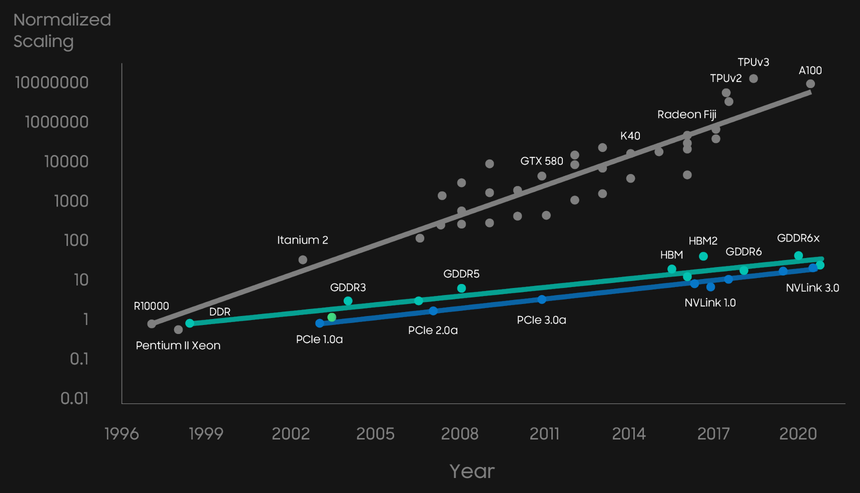
- PIM has been proposed to improve performance of bandwidth-sensitive workloads and increase energy efficiency by reducing data movement between computing and memory

AI and Memory Wall

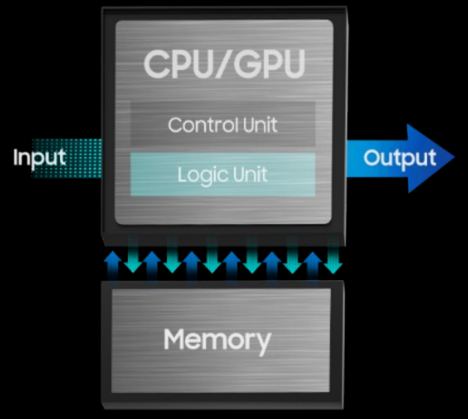


Transformer Size: 240x/2yrs
AI HW Memory: 2x/2yrs

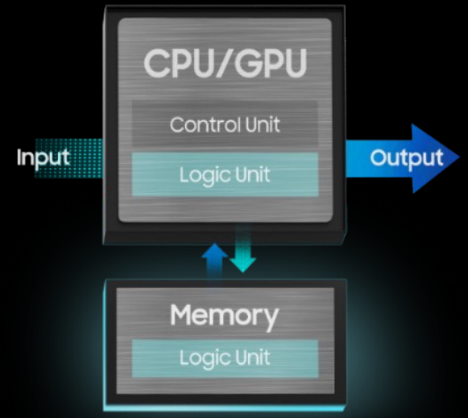
Scaling of Peak Hardware FLOPS, and Memory /Interconnect Bandwidth



HW FLOPS: 90,000x / 20 yrs (3.1x/2yrs)
DRAM BW: 30x /20 yrs (1.4x /2yrs)
Interconnect BW: 30x / 20 yrs (1.4x /2yrs)



Von Neumann



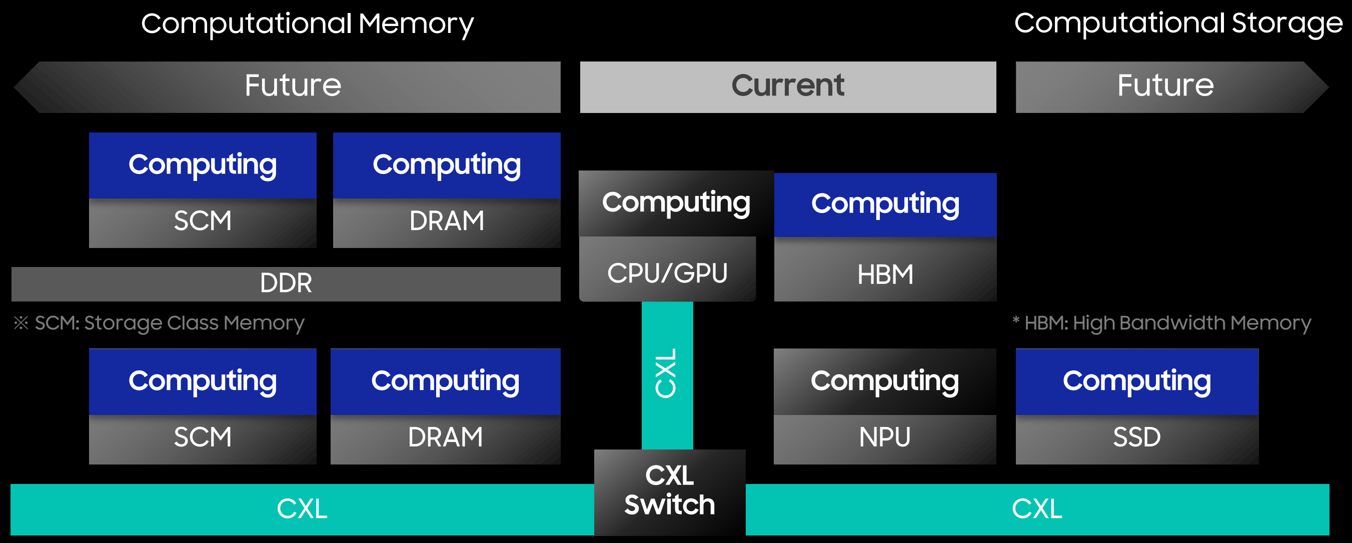
PIM

Source: https://daydaynews.cc/en/science/the-biggest-obstacle-to-ai-training-is-not-computing-power.html

Processing in Memory (PIM)

Future: Computing in Everywhere

- AI and Simulation have been major single large jobs and "Memory Wall" remains a critical issue
- Computation-enabled memory and storage are good approaches for breaking the "Memory Wall",
  but they increase the system complexity
- Need to choose the optimal combination of data movement and job migration
  → Job and Data Co-location problem
- We believe that new programming model and system SW such as OS will solve the problem



Panel discussion agenda

- How will the future computing machines be in terms of data-centric applications such as AI, Big-Data and HPC?
- Can you enumerate 1 or 2 important challenges to be faced by applications for PIM/PNM?
- What are the major challenges or missing features in the current programming models and runtime systems for PIM/PNM?
- What are the major challenges or missing features in the memory pool solutions and what will be needed to prepare for it?
- Can you find any solution to the challenges described before?

IBM

Paul G. Crumley

Senior Technical Staff Member
IBM Research

Paul G Crumley, a Senior Technical Staff Member at IBM Research, enjoys creating systems to solve problems beyond the reach of current technology.

Paul's current project integrates secure, compliant AI capabilities with enterprise Hybrid Cloud allowing clients to extract new business value from their data.

Paul's previous work includes the design and construction of distributed, and high-performance computing systems at CMU, Transarc, and IBM Research. Projects include The Andrew Project at CMU, ASCI White, IBM Global Storage Architecture, Blue Gene Supercomputers, IBM Cloud, and IBM Cognitive Systems.

Paul has managed data centers, and brings his first-hand knowledge of these environments, combined with experience of automation and robustness, to the design of AI for Hybrid Cloud infrastructure.

Position Statement

- Existing systems “waste” energy on work not directly used to solve the problem
- New technology allows us to create domain-specific chips and connect them securely with fabrics such as CXL
- We need new, open ecosystems which define industry standards for interoperability and tools to automate the end-to-end processes used to create the components and assemble systems
- These technologies and tools will allow us to create systems that are more performant and efficient by moving work close to the data and communication paths

AMD

Nuwan Jayasena

Fellow
AMD Research

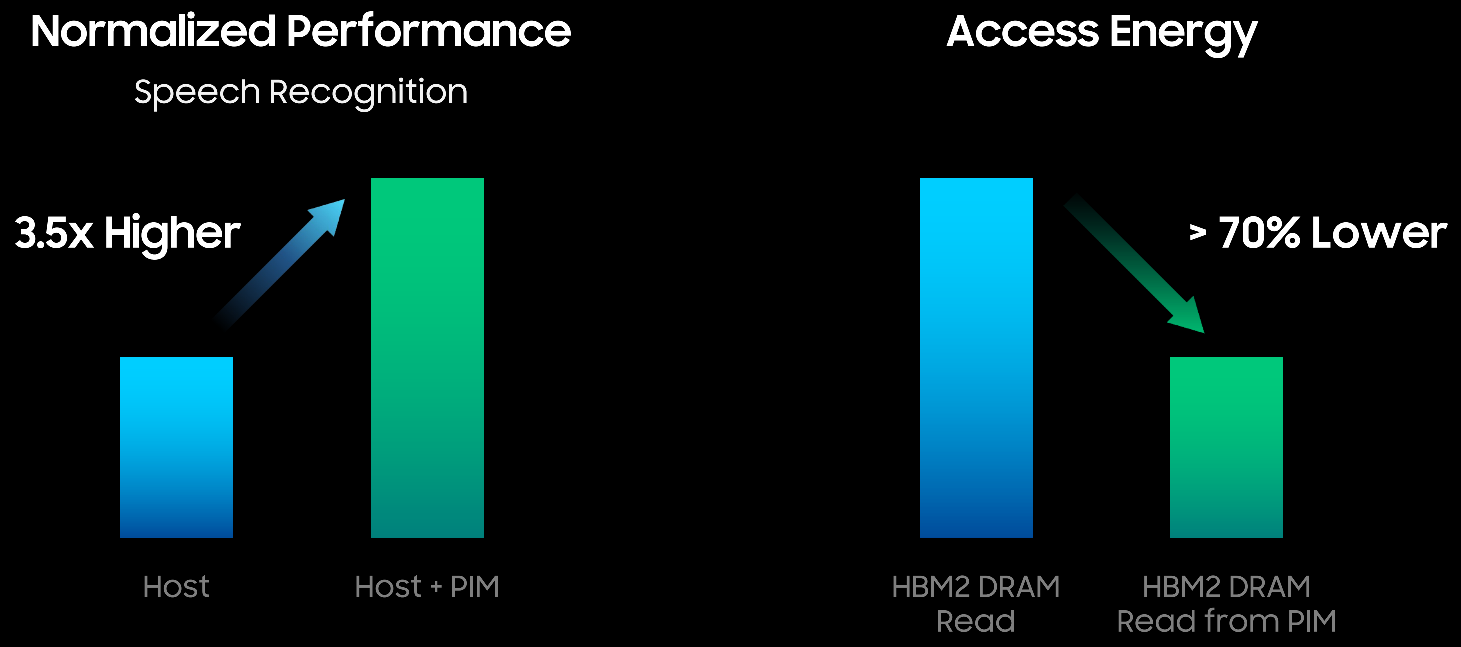
Nuwan Jayasena is a Fellow at AMD Research, and leads a team exploring hardware support, software enablement, and application adaptation for processing in memory. His broader interests include memory system architecture, accelerator-based computing, and machine learning.

Nuwan holds an M.S. and a Ph.D. in Electrical Engineering from Stanford University and a B.S. from the University of Southern California. He is an inventor of over 70 US patents, an author of over 30 peer-reviewed publications, and a Senior Member of the IEEE.

Prior to AMD, Nuwan was a processor architect at Nvidia Corp. and at Stream Processors, Inc.

The Promise of PIM

- Processing In Memory (PIM) significant performance and energy benefits
  - Example: HBM2-PIM (J.H. KIM et al., Hot Chips 2021)



- Energy efficiency may be the biggest benefit of PIM as more systems become power/energy-constrained

The Road to Processing in/near Memory Adoption

Workloads: What are the killer applications?

We may need to rethink workloads at the algorithm level to fully benefit from PIM

PIM will likely enable new opportunities/capabilities we are not exploiting today

Programmability: Few Developers want to deal with yet another class of accelerators

Early work in libraries and frameworks that can automatically target PIM/PNM is promising

System enablement: Resource management for in/near-memory computation is nascent

Memory managers must become aware of physical memory organization

Support for isolation, virtualization, security, RAS etc. is needed for broad adoption

Hardware nimbleness: PIM/PNM must keep up with fast-evolving workload landscape

Keep up with evolving numerical formats (FP16, BF16, FP8 (multiple), block & hierarchical formats, etc.)

Cater to conflicting capacity, bandwidth, latency requirements

OAK RIDGE
National Laboratory

Seyong lee

Senior R&D Staff
Oak Ridge National Laboratory

Seyong Lee is a Senior R&D Staff in the Computer Science and Mathematics Division at Oak Ridge National Laboratory (ORNL). His research interests include parallel programming and performance optimization in heterogeneous computing environments, program analysis, and optimizing compilers. He received his PhD in Electrical and Computer Engineering from Purdue University, USA.

He served as a program committee/guest editor/external reviewer for various conferences, journals, and research proposals. His paper on SC10 won the best student paper award, and his paper on PPoPP09 was selected as the most cited paper among all papers published in PPoPP between 2009 and 2014.

He received the IEEE Computer Society TCHPC Award for Excellence for Early Career Researchers in High Performance Computing at SC16 and served as an award committee member for 2017 IEEE CS TCHPC Award.

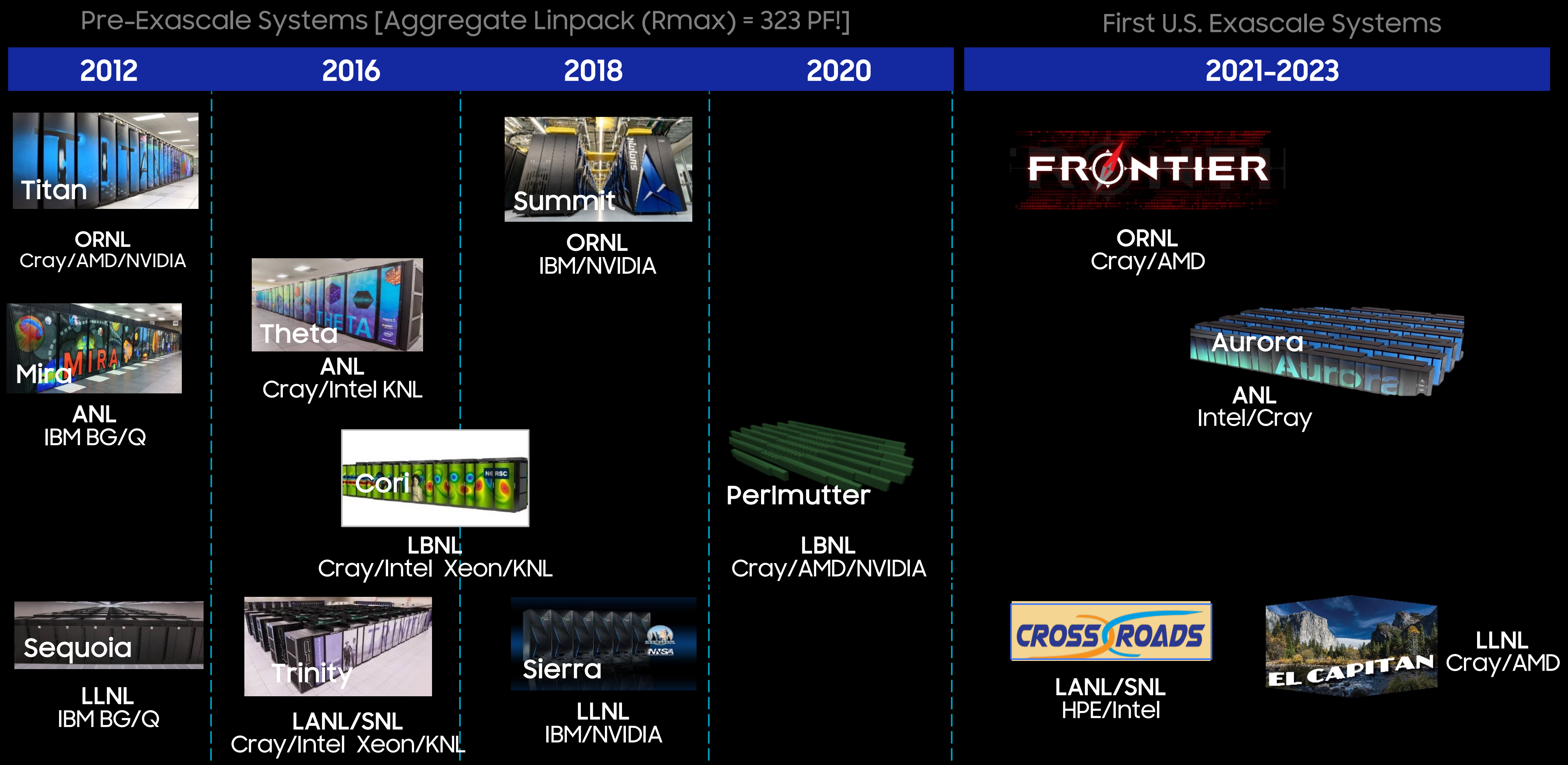
Position Statement

- Processing In Memory (PIM) can serve as an energy-efficient solution to tackle the data movement bottleneck caused by the memory wall problem.
- However, for wider adaption of PIM, several key challenges such as programmability, performance portability, and interoperability should be addressed.

Department of Energy (DOE) Roadmap to Exascale Systems

Oak Ridge National Laboratory

An impressive, product lineup of accelerated node systems supporting DOE's mission



Oak Ridge National Laboratory

Programming for Extremely Heterogeneous Architectures

ORNL Experimental Computing Laboratory (ExCL) systems (https://excl.ornl.gov/)

Systems	Snapdragon	Jetson	Zynq	DGX		Oswald		Summit	Frontier	
CPU	ARM	ARM	ARM	Intel	Intel	Intel	Intel	IBM	AMD	
GPU	Qualcomm	NVIDIA		NV	NV	NV	NV	NVIDIA	AMD	AMD
FPGA			Xilinx			Intel				
DSP	Qualcomm									

Snapdragon Host

OpenMP + OpenCL + Hexagon

OpenMP Kernel

OpenCL Kernel

Hexagon Kernel

Summit Host

OpenMP + CUDA

PIM Host

OpenMP Kernel

CUDA Kernel

PIM Kernel

Programming Model Choices for Extremely Heterogeneous Computing

Approaches used in DOE Exascale Computing Project (ECP)*

Device-specific Kernel Programming

- Isolate the computationally-intensive parts of the code into CUDA/HIP kernels
- Refactoring the code to work well with the device is the majority of effort.

Directive-based Programming

- Offload computationally-intensive parts to devices using OpenACC or OpenMP directive compilers.
- Most common portability system for Fortran codes.

C++ Abstractions

- Fully abstract loop execution and data management using advanced C++ features.
- Kokkos and RAJA developed by NNSA in response to increasing hardware diversity.

Domain-specific libraries and frameworks

- Design application with a specific motif to use common software components.
- Depend on co-design code (e.g., CEED, AMReX) to implement key functions on devices

* https://www.exascaleproject.org

SMART® Modular Technologies

Andy Mills

Senior Director,
Advanced Product Development
SMART Modular Technologies

Andy Mills is the Sr. Director of Advanced Product Development at SMART Modular Technologies, where he leads the development of next generation CXL memory and data center storage solutions.

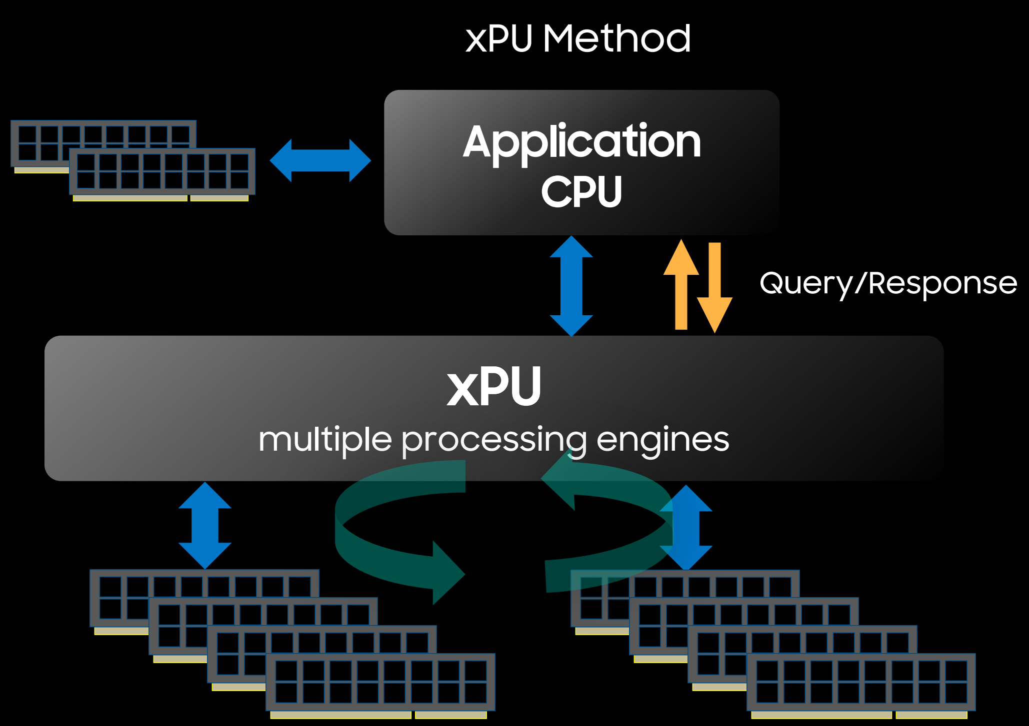
Prior to joining SMART, Andy was CEO/co-founder of Enmotus, Inc where he co-developed and led the development of intelligent storage and memory tiering solutions for data center and high end PCs utilizing advanced learning algorithms driven by real time intelligent workload analysis.

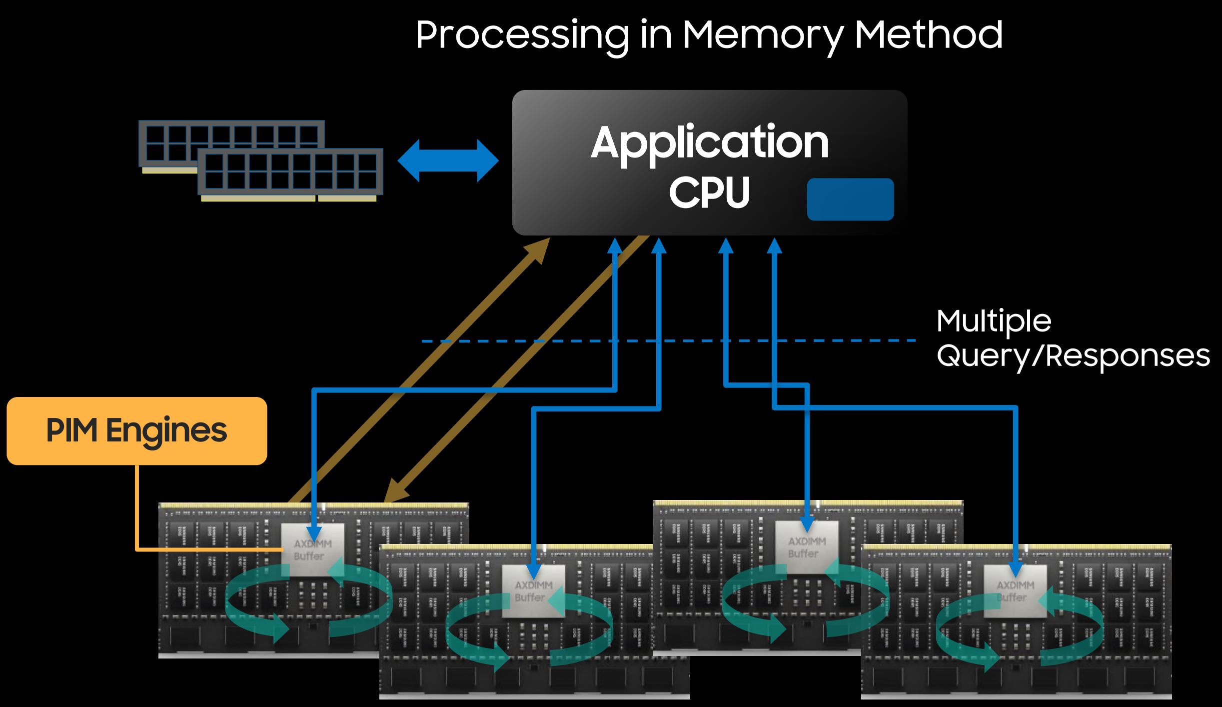
He has more than 30 years industry experience in software development, systems architecture, networking, storage and semiconductor development, plus held various management and technical lead positions at DotHill Systems, NetCell Corporation, TDK Semi and AMD. Andy received his Masters in Electronic Engineering from Bangor University in the UK.

Position Statement

- CXL will be key enabler for PIM.
- Many of today's parallel bus DIMM approaches to PIM face scale challenges as data sets continue to grow in complexity in size. CXL offers a cleaner way to add and scale memory and computational devices.

Offloading Memory Intensive Applications



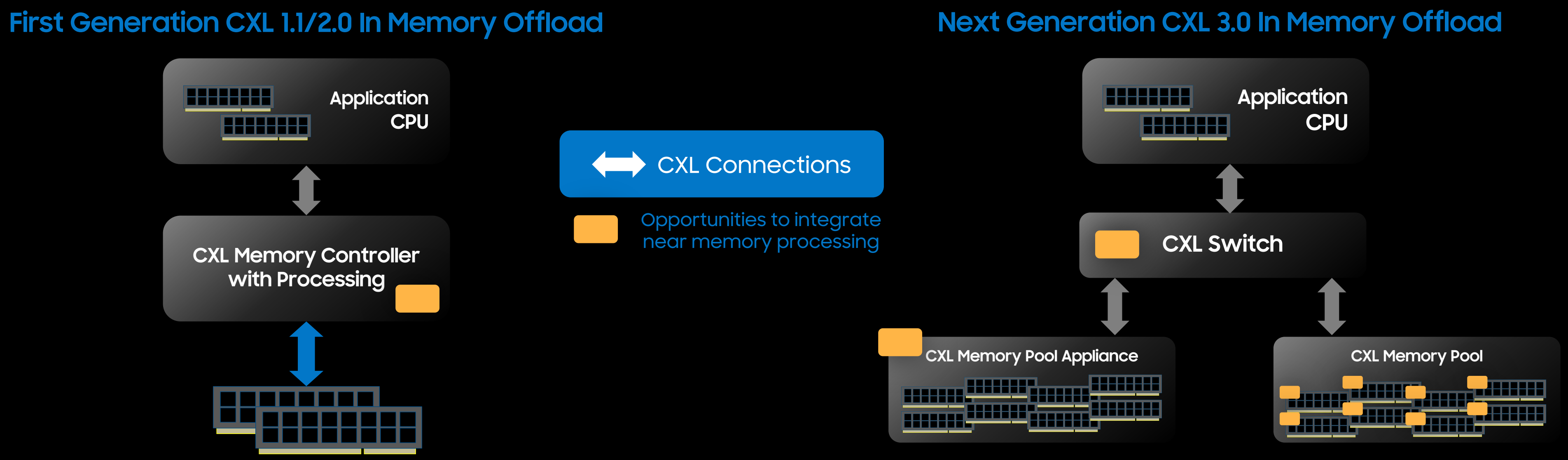


- Application only need to be aware of total size of dedicated xPU memory
- Application impact low as only one offload engine to interface with
- Conventional memory but uses more energy than we'd like as xPUs are power hungry in themselves
- Somewhat scalable by adding more xPU/memory sub-systems

- Opportunity to downsize CPU/memory
- Application most likely needs to align or change it's data set sizes to match the size of the individual module
- Application impact high as we need to talk to "n" different engines
- Scaling limits due to memory channel limits (for DIMM based)

CXL Accelerators, CXL Memory and Benefits to PIM

- CXL – Compute eXpress Link
  - Industry standard built ontop of the highly prevalent PCIe bus (Gen 5 and newer)
  - Introduced in new CPU platforms from both AMD and Intel in 2023
- An industry standard way to decouple memory from CPUs and share memory between multiple CPUs
- Designed for accelerator cache coherent (cxl.cache), memory (cxl.mem) and IO (cxl.io) configurations
- Overcomes memory channel limitations of single/dual/quad CPU systems



Elastics.
CLOUD

George Apostol

Founder & CEO
Elastic.cloud

George has over 35 years of experience designing system-on-chip (SoC), hardware, software, and systems. He holds several patents for interconnect and interface design and has led technology organizations developing products for several markets.

He has held leadership and executive roles at Xerox/PARC, Sun, SGI, LSI Logic, Exar, and Samsung as well as start-ups including TiVo, BRECIS, Audience and BAYIS.

As VP of Engineering and CTO at PLX Technology, George drove the development of PCI Express switches, which are broadly used in the market today. He received his bachelor's degree in Electrical Engineering from the Massachusetts Institute of Technology.

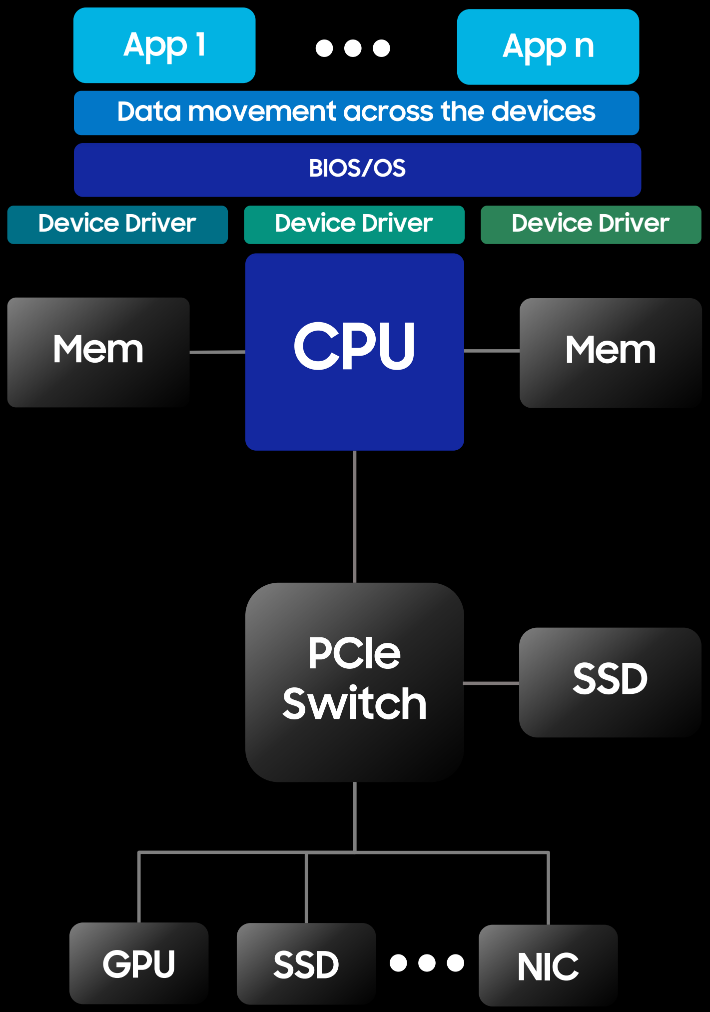
Position Statement

Elastics.cloud

- Elastics.cloud is delivering the key technology to revolutionize system-level performance in data centers.
- Leveraging CXL to provide distributed control and efficient resource management, Elastics.cloud's switch SoC (SSoC) solution allows flexibility in the system architecture surrounding PIM (processing in memory) devices, computational storage devices, and also fully enables PNM (processing near memory) architectures by allowing any individual processing element to access any memory or storage element with only two hops through the switch, avoiding CPU bottlenecks as seen in today's Von Neumann architectures.

Improved System Performance with CXL

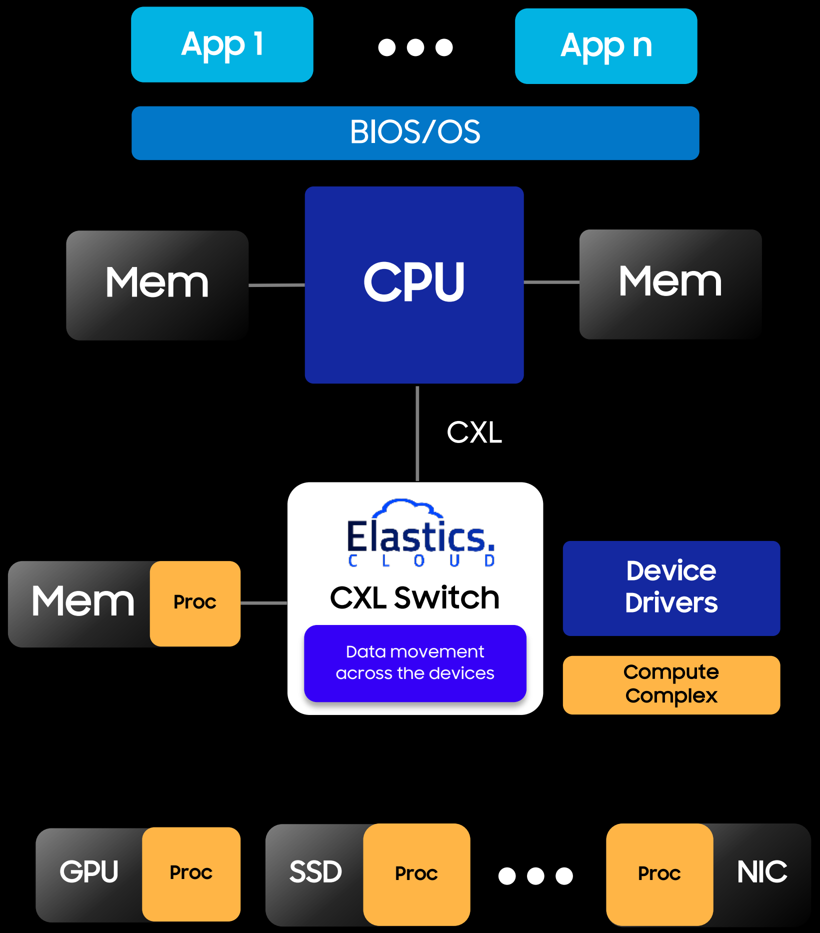
PCIe Architecture – Current



Enablement of PIM/PNM with CXL Switch:
Free up cycles on CPU with data movement and device driver layers moved to the switch

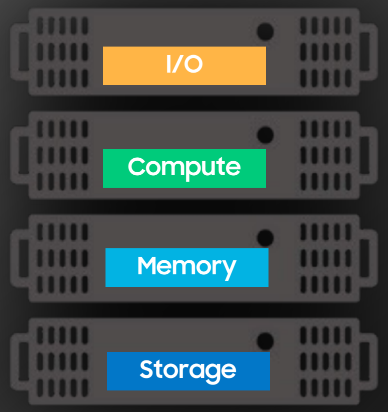
Increased system architecture flexibility:
Compute, memory, storage elements all attached through SSoC

CXL Architecture



Elastic.cloud Mission

Future Rack-Scale Architecture



Composable set of pooled, shared and disaggregated compute, network, memory, & storage resources

- Improved utilization of components
- Flexibility to right-size resources to match workloads
- Component expansion with high-speed, low-latency interconnect
- Multi-host, shared component enablement
- Intelligent buffering and traffic management
- Software-managed orchestration

Applications

Microservices, containers, server storage, virtual networks, NOS, programmable silicon

Low-latency, distributed control switch fabric

Hardware attributes exposed to provisioning management layer

ML/AI

Service Orchestration | **Infrastructure Orchestration** | **Network Orchestration**

- Leveraging the Compute Express Link (CXL) interconnect to deliver new composable, heterogeneous architectures
- CXL based components intelligently composed with software orchestration to optimize performance and TCO

Thank You